

AIHUI, China

WMO: 504680

Lat: 50.2481N

Lon: 127.4619E

Elev: 545

StdP: 14.41

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-23.6	-33.4	0.8	-26.7	-29.9	1.0	-22.8	17.9	4.5	16.1	1.8	3.2	320	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	87.6	68.8	84.4	68.2	81.6	67.2	73.5	81.8	71.8	79.6	70.1	77.7	8.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.0	116.6	77.4	69.3	109.8	75.8	67.6	103.4	74.3	37.6	82.2	36.1	79.7	34.6	77.6	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.7	13.6	-30.6	93.3	4.9	3.3	-34.2	95.7	-37.1	97.6	-39.8	99.5	-43.4	101.8		
(5)			-30.8	76.4	4.9	2.4	-34.4	78.1	-37.2	79.5	-40.0	80.8	-43.5	82.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	-6.0	2.1	18.9	39.6	54.5	66.0	71.0	66.9	54.5	37.0	13.1	-4.4		
	DBStd	28.80	8.61	8.97	10.98	8.16	7.56	6.33	4.65	5.43	7.39	8.72	12.09	8.28		
	HDD50	7647	1737	1340	964	328	38	0	0	1	38	410	1106	1685		
	HDD65	11496	2202	1760	1429	763	337	63	7	41	321	867	1556	2150		
	CDD50	2031	0	0	0	16	177	480	652	524	174	8	0	0		
	CDD65	406	0	0	0	1	12	92	194	100	7	0	0	0		
	CDH74	3498	0	0	0	17	230	969	1453	735	92	2	0	0		
	CDH80	968	0	0	0	3	61	318	414	163	9	0	0	0		
	WSAvg	5.7	5.1	5.4	6.6	7.4	6.9	5.3	4.8	4.8	5.4	5.9	5.4	5.0		
	PrecAvg	20.1	0.1	0.1	0.2	0.9	1.7	3.3	5.1	4.9	2.4	0.8	0.3	0.2		
(15) Precipitation	PrecMax	30.9	0.2	0.4	0.7	5.0	6.0	5.7	12.1	9.6	4.6	2.7	1.1	0.5		
	PrecMin	13.5	0.0	0.0	0.0	0.1	0.2	0.7	0.6	1.4	0.4	0.0	0.0	0.0		
	PrecStd	4.0	0.1	0.1	0.2	0.9	1.2	1.4	2.9	2.0	1.3	0.6	0.3	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	33.3	52.4	75.9	86.4	92.8	92.5	87.7	80.9	68.2	47.5	21.1		
		MCWB	15.8	26.7	39.0	51.6	59.2	68.0	72.0	71.0	64.8	51.0	38.6	18.8		
	2%	DB	13.8	26.7	45.9	66.0	80.4	87.5	87.9	84.2	76.6	60.5	39.9	16.1		
		MCWB	11.1	22.2	35.5	47.3	57.6	66.8	71.4	70.5	62.0	48.0	33.4	14.2		
	5%	DB	10.2	21.0	40.2	60.1	75.0	83.3	84.5	81.4	72.6	55.3	34.5	11.5		
		MCWB	8.0	17.3	32.0	44.6	55.1	65.1	70.4	68.9	60.0	44.8	29.8	9.5		
	10%	DB	6.7	16.4	35.8	55.2	70.3	79.6	81.6	78.2	68.3	51.0	30.1	7.9		
		MCWB	4.9	13.4	29.3	42.0	53.2	64.2	69.4	67.4	57.5	42.0	26.6	6.2		
	0.4%	WB	15.8	27.1	40.2	53.6	62.1	73.3	76.5	75.3	68.1	54.7	38.7	19.1		
		MCDB	18.5	32.4	50.6	71.9	77.1	86.2	85.1	83.2	76.1	64.3	46.2	21.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.1	22.4	35.9	48.7	59.4	70.0	74.1	72.7	64.6	49.4	33.7	14.0		
		MCDB	13.7	26.4	44.9	63.7	75.2	81.5	82.6	80.2	72.3	57.5	39.3	16.2		
	5%	WB	8.0	17.4	32.4	45.5	57.3	67.7	72.7	70.8	62.0	45.9	30.5	9.6		
		MCDB	10.0	20.8	39.3	58.3	71.1	78.1	80.4	78.1	69.8	53.9	34.2	11.4		
	10%	WB	4.9	13.4	29.7	42.6	55.0	65.8	71.2	69.0	59.4	42.9	26.5	6.2		
		MCDB	6.6	16.3	35.6	53.4	67.3	76.3	78.6	75.8	66.0	50.2	30.1	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	21.1	20.6	20.7	21.7	19.9	16.9	17.7	21.0	18.7	16.9	15.8		
		MCDBR	19.7	23.6	23.4	29.9	30.5	27.3	22.0	22.6	27.0	26.3	20.4	17.5		
		MCWBR	17.6	19.7	15.3	16.3	13.2	10.8	8.9	10.3	13.6	15.3	14.9	15.5		
		MCDBR	19.5	23.6	21.9	27.4	25.7	21.8	17.5	18.3	22.2	23.4	19.9	17.2		
	5% WB	MCWBR	17.4	19.8	14.7	15.3	12.1	10.1	8.7	9.7	12.4	14.7	14.8	15.4		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.302	0.374	0.507	0.566	0.501	0.484	0.431	0.393	0.387	0.319	0.279		
	taud		2.161	2.090	1.955	1.791	1.733	2.003	2.107	2.240	2.263	2.117	2.151	2.118		
	Ebn at Noon		238	259	256	232	225	243	246	254	251	222	213	212		
	Edh at Noon		24	34	47	62	69	53	47	40	35	33	25	22		
(44) All-Sky Solar Radiation	RadAvg		532	915	1401	1585	1724	1853	1717	1526	1227	816	525	413		
	RadStd		17	31	89	168	143	236	144	107	103	49	31	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page